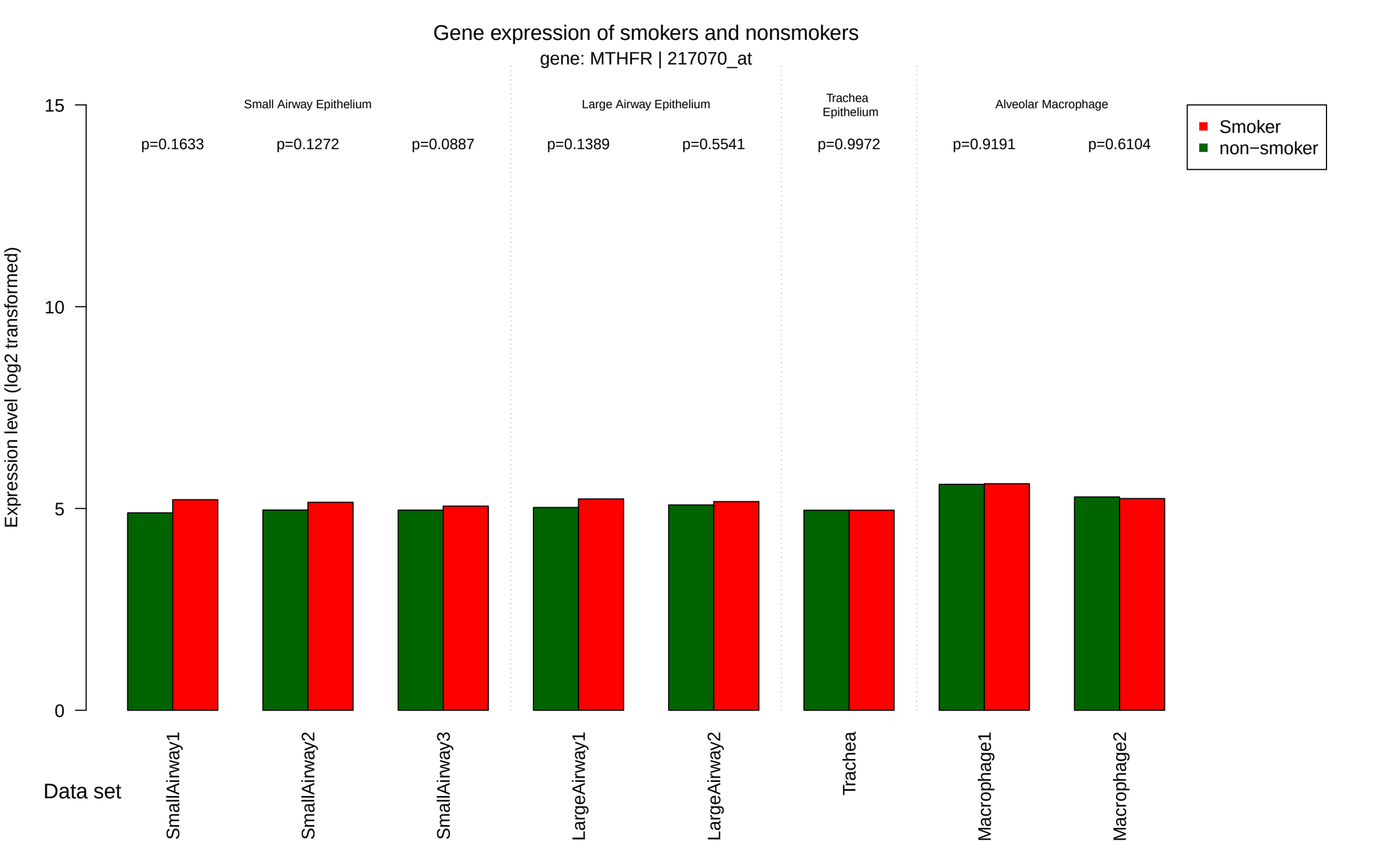
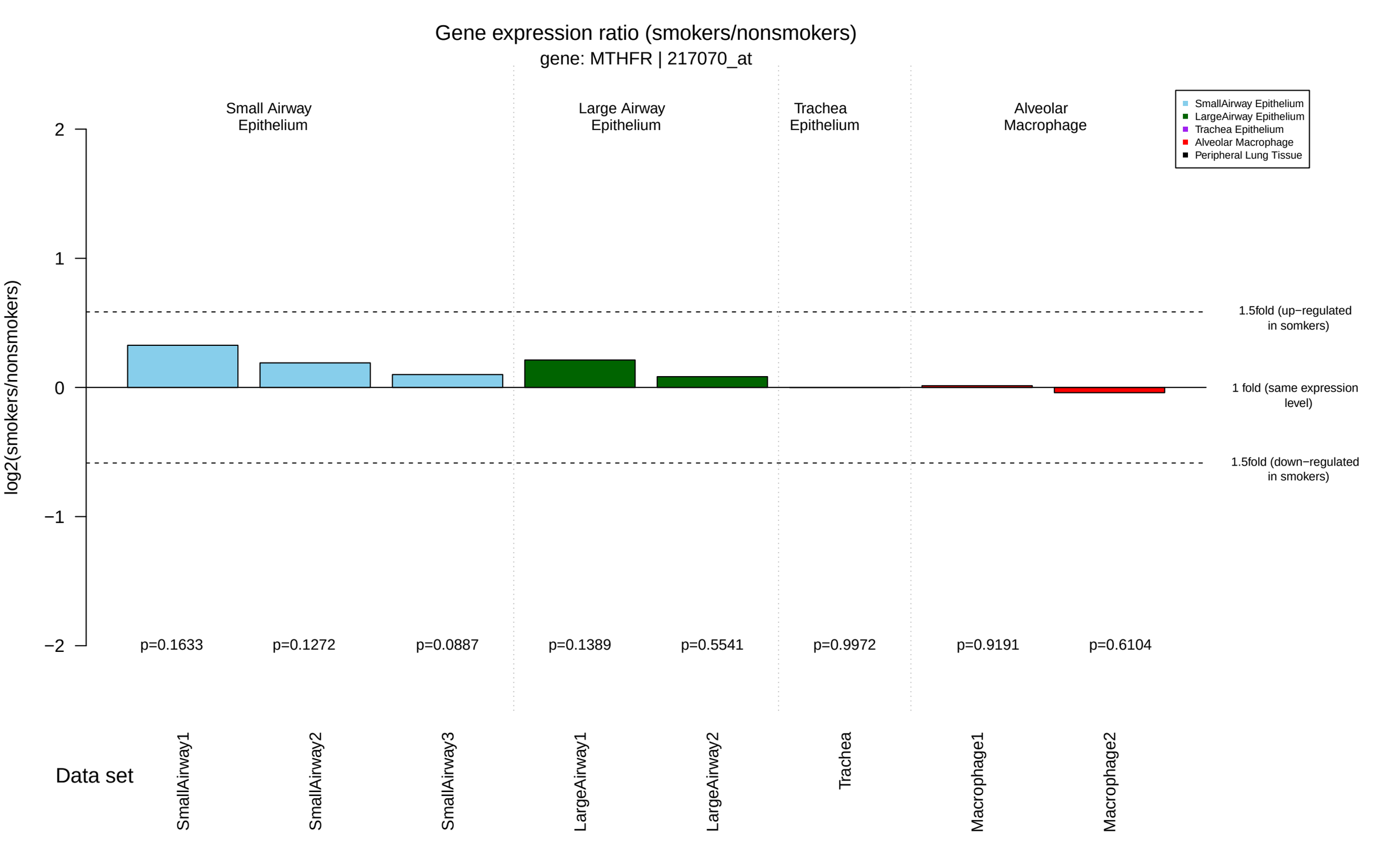
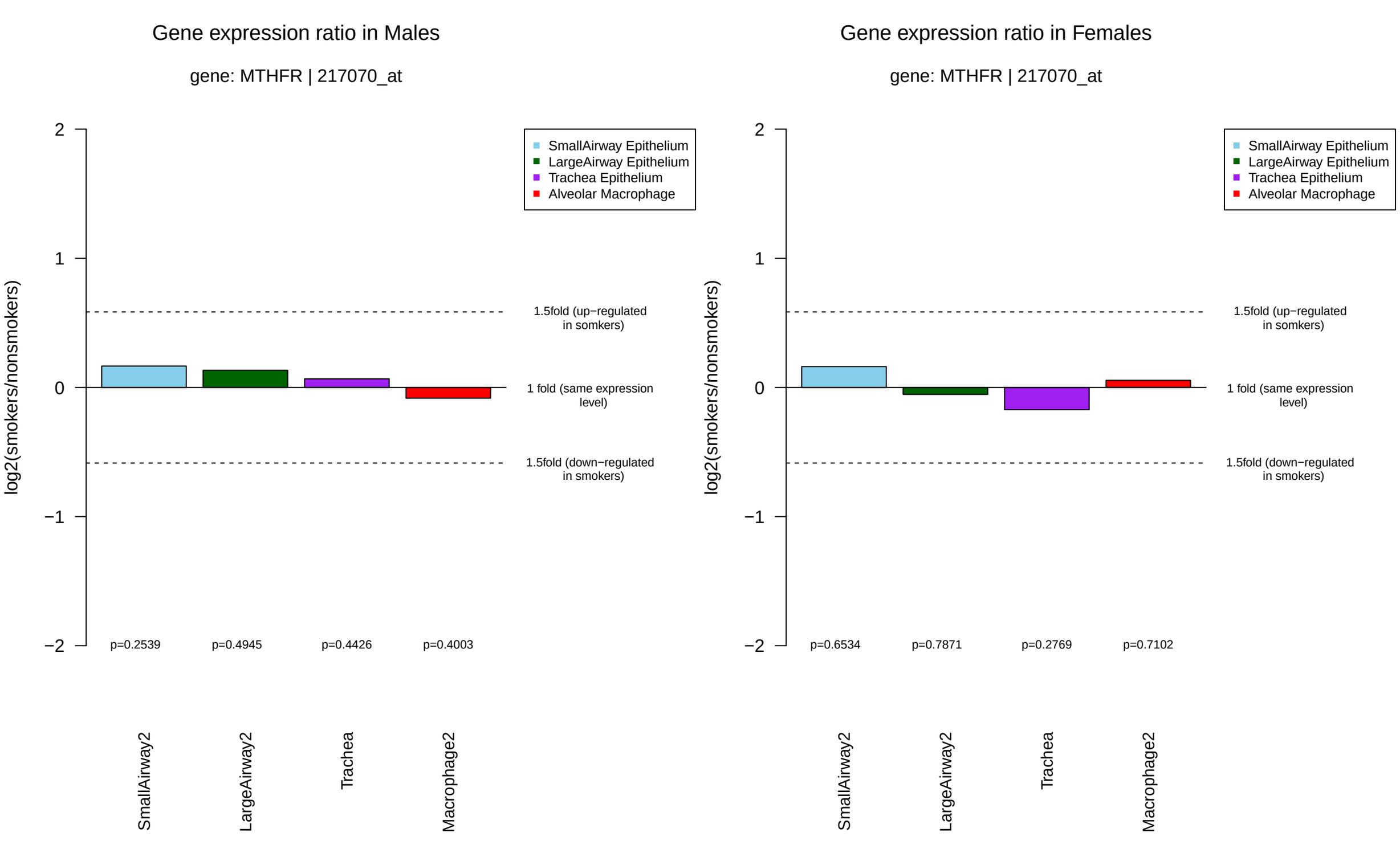
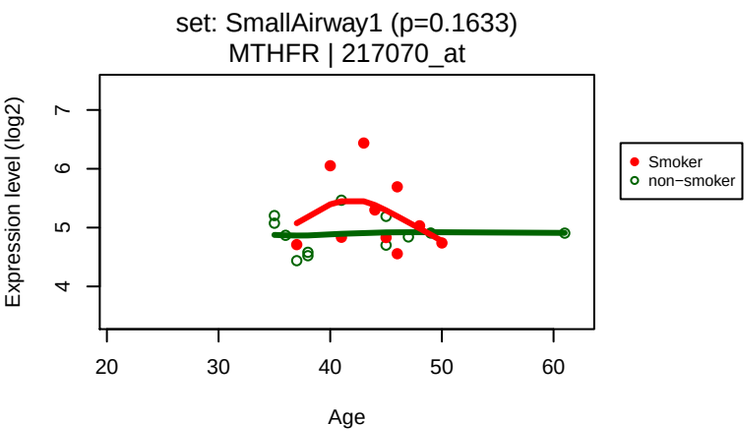
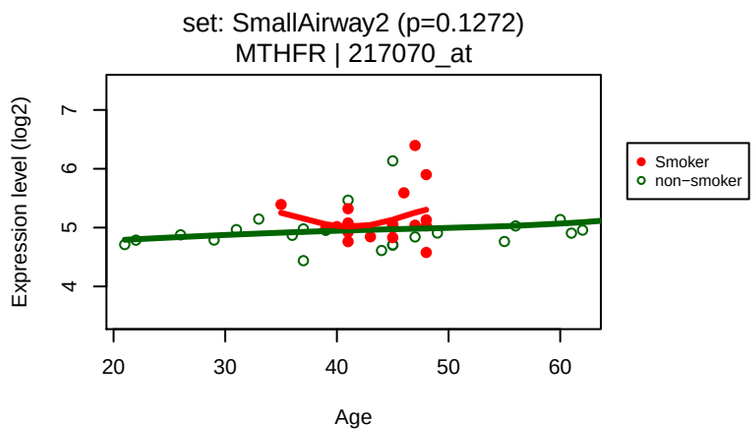
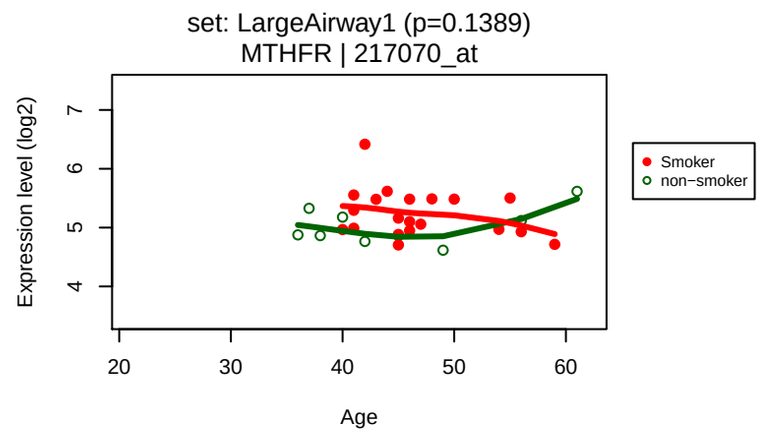

Gene expression of smokers and nonsmokers
gene: MTHFR | 217070_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1633
p=0.1272
p=0.0887
p=0.1389
p=0.5541
p=0.9972
p=0.9191
p=0.6104
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

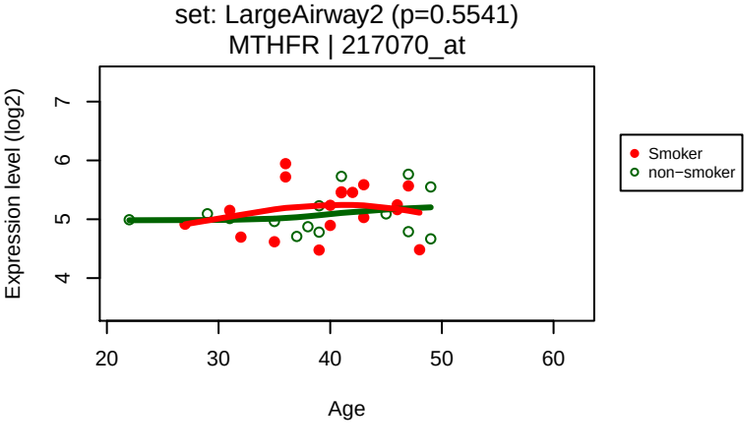

Gene expression ratio (smokers/nonsmokers)
gene: MTHFR | 217070_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1633
p=0.1272
p=0.0887
p=0.1389
p=0.5541
p=0.9972
p=0.9191
p=0.6104
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

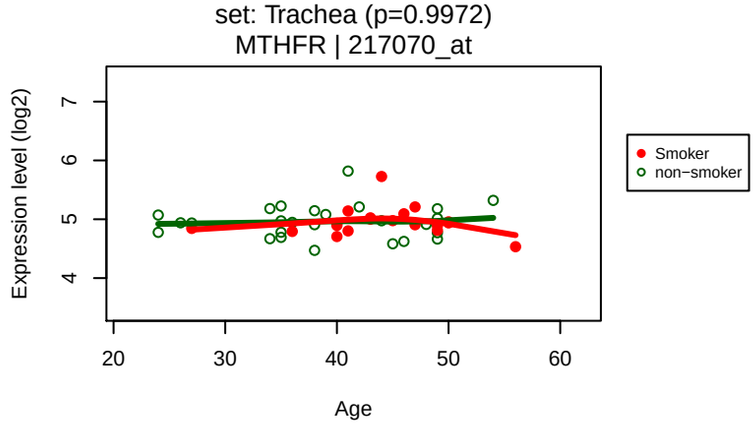

Gene expression ratio in Males
gene: MTHFR | 217070_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2539
p=0.4945
p=0.4426
p=0.4003
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MTHFR | 217070_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6534
p=0.7871
p=0.2769
p=0.7102
SmallAirway2
LargeAirway2
Trachea
Macrophage2

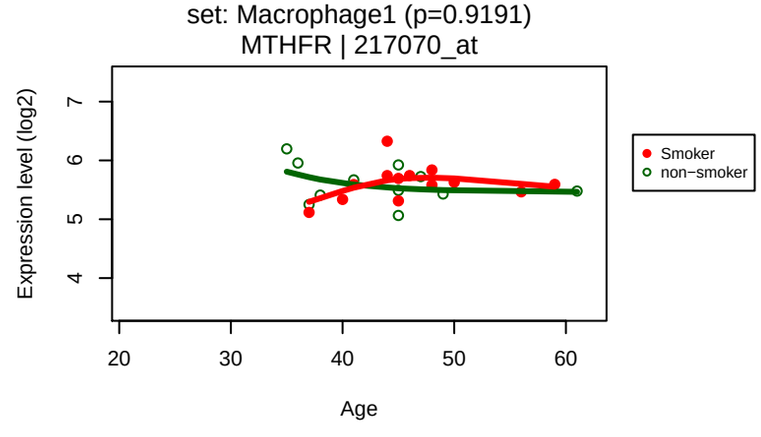

set: SmallAirway1 (p=0.1633)
MTHFR | 217070_at
Expression level (log2)
Age
Smoker
non−smoker

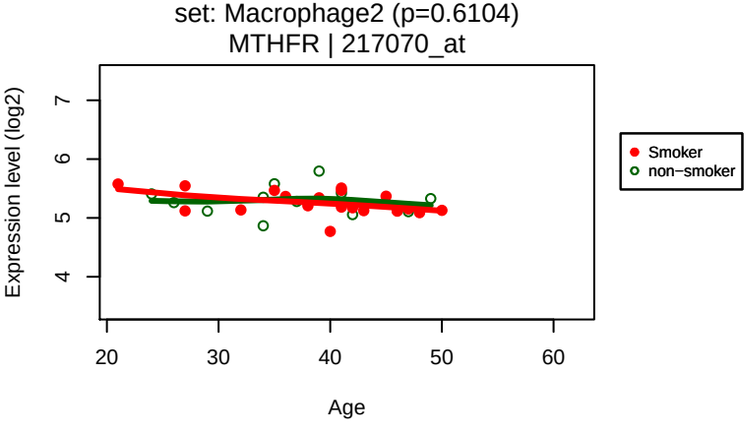

set: SmallAirway2 (p=0.1272)
MTHFR | 217070_at
Expression level (log2)
Age
Smoker
non−smoker

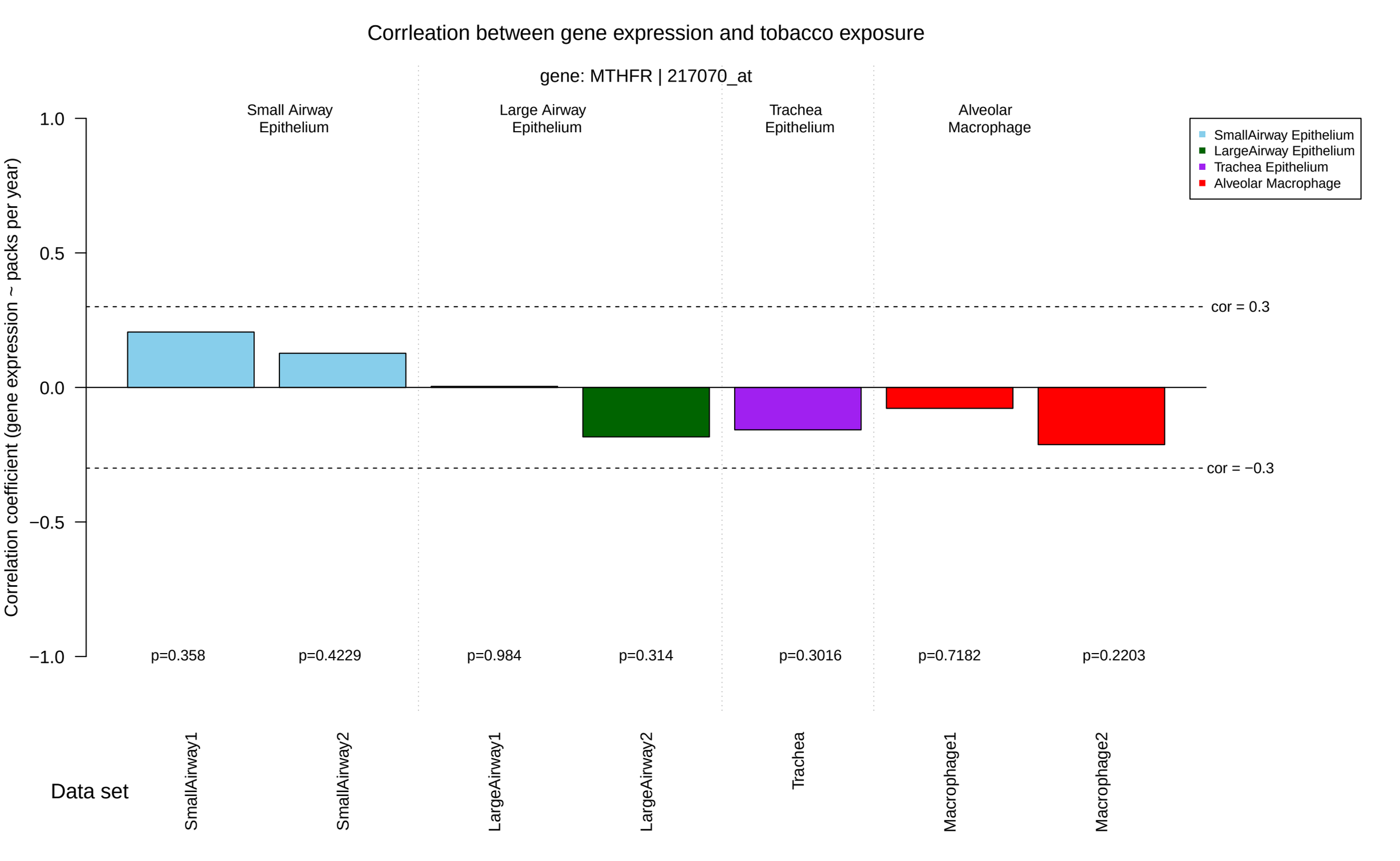

set: LargeAirway1 (p=0.1389)
MTHFR | 217070_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5541)
MTHFR | 217070_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9972)
MTHFR | 217070_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9191)
MTHFR | 217070_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6104)
MTHFR | 217070_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MTHFR | 217070_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.358
p=0.4229
p=0.984
p=0.314
p=0.3016
p=0.7182
p=0.2203
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2